

AAMIR GULZAR

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SUMMARY

AI Engineer specializing in production Generative AI and agentic systems: RAG architectures, real-time voice pipelines, and enterprise automation, backed by a Q1-published first-author research background in medical AI. Experienced across the full stack from data engineering and model fine-tuning to production deployment and MLOps.

TECHNICAL SKILLS

Agentic AI & RAG: LangChain, Pipecat, custom/deterministic agent orchestration, Prompt/Context Engineering; working knowledge of LangGraph, CrewAI, LlamaIndex, NeMo Guardrails.

Generative AI Engineering: LLM Fine-tuning (LoRA/QLoRA), n-bit Quantization, Foundation Models, Agentic RAG, MediaPipe on-device inference; working knowledge of vLLM, Diffusers, Flux, DreamBooth, Ragas & DeepEval, LangSmith.

Speech & Multimodal: Real-Time ASR/TTS/STT Pipelines, Whisper STT, Piper TTS, WebRTC, VLMs.

Cloud, DevOps & MLOps: Google Cloud (Vertex AI, BigQuery, Cloud Run, GCS), Azure, AWS (EC2), CI/CD, Docker, Kubernetes.

Data & Vector Databases: Qdrant, ChromaDB, PGVector, Vertex AI Vector Search, PostgreSQL, MongoDB, Snowflake, Data Pipelines.

Core Frameworks: PyTorch, TensorFlow, FastAPI, Streamlit, React, Python (Pandas, NumPy) and others.

PRODUCTION AI ENGINEERING

Independent AI Engineer

Jun 2026 – Present

Self-employed

Remote

- Built a production-scale Multi-Agentic RAG system for a confidential US (California State) client, using Vertex AI Vector Search (3072-dim), Gemini Embedding 001, and Gemini 2.5 Flash on GCP Cloud Run.
- Engineered incremental manifest-based ingestion (skipped on re-runs) and a data-repair pipeline correcting thousands of documents with zero data loss.
- Achieved sub-500ms p99 query latency with low infrastructure cost through semantic chunking and metadata-first indexing.
- Orchestrated ingestion and classification via Kubeflow v2 pipelines on Vertex AI, using a shared circuit-breaker rate limiter and VLM-assisted document classification to keep the knowledge base consistent with zero downtime.

AI Engineer

Nov 2025 – Jun 2026

TalentBridge Oy (Client: Neste)

Espoo, Finland (Remote)

- Developed an automated contract generation system for Neste using Vertex AI Pipelines, Snowflake, and Google Docs/Drive integration, achieving 90%+ automated accuracy and 98%+ field accuracy - scaling operations roughly 20x.
- Built SUSAN, a voice-enabled, RAG-powered AI assistant (SUSAN-GPT) delivering career, business, and integration guidance to jobseekers, students, and entrepreneurs in Finland (susan.fi).
- Developed Staffin, an AI-powered resume-matching and job discovery platform, and a companion multi-agent AI interview system for automated candidate screening (staffin.fi).

AI Product Advisor (R&D)

Jan 2025 – Jun 2025

APIMatic

New Zealand (Remote)

- Led development of an R&D VSCode extension for AI Copilot, optimizing code generation for API integrations.
- Engineered a reactive UI (React + Vite) with a FastAPI backend backed by Azure Storage and Entra ID.

ML Consultant

Machine Learning Engineer

PlanetBeyond

Mar 2024 – Oct 2024

Aug 2023 – Mar 2024

Islamabad, Pakistan

- Engineered a real-time autonomous AI voice caller using Whisper STT, Piper TTS, WebRTC, and Docker on AWS.
- Built a multi-format financial RAG QA application for SEC filings using semantic retrieval, multilingual pre-indexing, and ROUGE-scored summarization.
- Conducted on-device optimization of open-source LLMs via MediaPipe (Gemma 2 2B, INT8), reaching sub-2s first-token latency on Snapdragon 680 with 95%+ quality retention.
- Fine-tuned and quantized (8-bit) Gemma-2 and Llama-3.1 for a conversational AI engine.

R&D, LEADERSHIP & TRAINING

Lecturer

Lab Instructor

National University of Computer and Emerging Sciences (FAST-NUCES)

Mar 2024 – Present

Aug 2021 – Mar 2024

Islamabad, Pakistan

- Delivered core academic lectures in AI, Machine Learning, and Programming; supervised ML Final Year Projects.
- Conducted specialized corporate AI training workshops for the National Police Academy (NPA) and NAVTTC.

AI Research Scientist / Team Lead

Research Assistant

DataInsight Research Lab

Jul 2023 – Oct 2025

Jul 2022 – Jul 2023

International Collaboration

- Benchmarked pathology foundation models for tumor classification on TCGA whole slide images (published, see below).
- Led a multimodal spatial transcriptomics integration study for ecotype and liver-metastasis prediction in pancreatic cancer, including a pre-registered, confound-audited validation protocol.
- Mentored and supervised three cohorts of AI interns on production and research-level lab projects.

PUBLICATIONS

- **Q1 Journal (2025):** "Benchmarking pathology foundation models for predicting microsatellite instability in colorectal cancer histopathology." *Computerized Medical Imaging and Graphics*.
- **Book Chapter (2024):** First author on a clinical comparative analysis of classical ML vs. Deep Learning for medical image analysis.

EDUCATION

Master of Science in Computer Science, FAST School of Computing (NUCES)

Bachelor of Science in Computer Science, PMAS Arid Agriculture University

Sep 2021 – Jan 2024

Aug 2017 – Jul 2021

LICENSES & CERTIFICATIONS

- **IBM:** Building AI Agents and Agentic Workflows (2025)
- Fundamentals of AI Agents using RAG & LangChain
- **Microsoft:** AI & ML Engineering (2025)
- **DeepLearning.AI & AWS:** Generative AI with Large Language Models